

JEE Mains 2019 Chapter wise Question Bank

Experimental Physics - Questions

Q1

A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is:

- (1) 2.0% (2) 2.5%
(3) 1.0% (4) 0.5%

9 Jan Morning

Q2

Surface of certain metal is first illuminated with light of wavelength $\lambda_1 = 350$ nm and then, by light of wavelength $\lambda_2 = 540$ nm. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of (2) The work function of the metal (in eV) is close to:

- (Energy of photon = $\frac{1240}{\lambda \text{ (in nm)}} \text{ eV}$)
(1) 1.8 (2) 2.5
(3) 5.6 (4) 1.4

9 Jan Morning

Q3

The magnetic field associated with a light wave is given at the origin by

$$B = B_0 [\sin(3.14 \times 10^7)ct + \sin(6.28 \times 10^7)ct].$$

If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of the photoelectrons?

$$(c = 3 \times 10^8 \text{ ms}^{-1}, h = 6.6 \times 10^{-34} \text{ J-s})$$

- (1) 6.82 eV (2) 12.5 eV
(3) 8.52 eV (4) 7.72 eV

9 Jan Evening

Q4

The pitch and the number of divisions, on the circular scale for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 division below the mean line.

The readings of the main scale and the circular scale, for a thin sheet, are 5.5 mm and 48 respectively, the thickness of the sheet is:

- (1) 5.755 mm (2) 5.950 mm
(3) 5.725 mm (4) 5.740 mm

9 Jan Evening

Q5

In an electron microscope, the resolution that can be achieved is of the order of the wavelength of electrons used. To resolve a width of 7.5×10^{-12} m, the minimum electron energy required is close to:

- (1) 500 keV (2) 100 keV
(3) 1 keV (4) 25 keV

10 Jan Morning

Q6

The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figures?

- (1) $4264 \pm 81 \text{ cm}^3$ (2) $4264.4 \pm 81.0 \text{ cm}^3$
(3) $4260 \pm 80 \text{ cm}^3$ (4) $4300 \pm 80 \text{ cm}^3$

10 Jan Evening

Q7

Experimental Physics

A metal plate of area $1 \times 10^{-4} \text{ m}^2$ is illuminated by a radiation of intensity 16 mW/m^2 . The work function of the metal is 5 eV . The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photo electrons per second and their maximum energy, respectively, will be: [$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$]

- (1) 10^{14} and 10 eV (2) 10^{12} and 5 eV
(3) 10^{11} and 5 eV (4) 10^{10} and 5 eV

10 Jan Evening

Q8

If the deBroglie wavelength of an electron is equal to 10^{-3} times the wavelength of a photon of frequency $6 \times 10^{14} \text{ Hz}$, then the speed of electron is equal to :

(Speed of light = $3 \times 10^8 \text{ m/s}$)

Planck's constant = $6.63 \times 10^{-34} \text{ J.s}$

Mass of electron = $9.1 \times 10^{-31} \text{ kg}$

- (1) $1.1 \times 10^6 \text{ m/s}$ (2) $1.7 \times 10^6 \text{ m/s}$
(3) $1.8 \times 10^6 \text{ m/s}$ (4) $1.45 \times 10^6 \text{ m/s}$

11 Jan Morning

Q9

In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm . The decrease in the stopping potential is close to :

$$\left(\frac{hc}{e} = 1240 \text{ nm-V} \right)$$

- (1) 0.5 V (2) 1.5 V
(3) 1.0 V (4) 2.0 V

11 Jan Evening

Q10

The least count of the main scale of a screw gauge is 1 mm . The minimum number of divisions on its circular scale required to measure $5 \mu\text{m}$ diameter of a wire is:

- (1) 50 (2) 200
(3) 100 (4) 500

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Q11

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A particle A of mass 'm' and charge 'q' is accelerated by a potential difference of 50 V . Another particle B of mass '4m' and charge 'q' is accelerated by a potential difference of

2500 V . The ratio of de-Broglie wavelength $\frac{\lambda_A}{\lambda_B}$ is

- (1) 10.00 (2) 0.07
(3) 14.14 (4) 4.47

12 Jan Morning

Q12

When a certain photosensitive surface is illuminated with monochromatic light of frequency ν , the stopping potential for the photo current is $-V_0/2$. When the surface is illuminated by monochromatic light of frequency $\nu/2$, the stopping potential is $-V_0$. The threshold frequency for photoelectric emission is :

- (1) $\frac{5\nu}{3}$ (2) $\frac{4}{3}\nu$
(3) 2ν (4) $\frac{3\nu}{2}$

12 Jan Evening

Q13

In a Frank-Hertz experiment, an electron of energy 5.6 eV passes through mercury vapour and emerges with an energy 0.7 eV . The minimum wavelength of photons emitted by mercury atoms is close to :

- (1) 1700 nm (2) 2020 nm
(3) 220 nm (4) 250 nm

12 Jan Evening

Q14

Two particles move at right angle to each other. Their de Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de Broglie wavelength λ , of the final particle, is given by :

- (1) $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$ (2) $\lambda = \sqrt{\lambda_1 \lambda_2}$
(3) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$ (4) $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

8 April Morning

Q15

Experimental Physics

In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to :

- (1) 0.7% (2) 0.2% (3) 3.5% (4) 6.8%

8 April Evening

Q16

In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is:

- (1) 0.01 kg/m^3 (2) 0.10 kg/m^3
(3) 0.31 kg/m^3 (4) 0.07 kg/m^3

9 April Morning

Q17

The electric field of light wave is given as

$$\vec{E} = 10^3 \cos$$

$$\left(\frac{2\pi x}{5 \times 10^{-7}} - 2\pi \times 6 \times 10^{14} t \right) \hat{x} \frac{N}{C}$$

This light falls on a metal plate of work function 2eV. The stopping potential of the photo-electrons is:

$$\text{Given, } E \text{ (in eV)} = \frac{12375}{\lambda \text{ (in } \text{\AA})}$$

- (1) 2.0 V (2) 0.72 V
(3) 0.48 V (4) 2.48 V

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Q18

A particle 'P' is formed due to a completely inelastic collision of particles 'x' and 'y' having de-Broglie wavelengths ' γ_x ' and ' γ_y ' respectively. If x and y were moving in opposite directions, then the de-Broglie wavelength of 'P' is:

- (1) $\frac{\gamma_x \gamma_y}{\gamma_x + \gamma_y}$ (2) $\frac{\gamma_x \gamma_y}{|\gamma_x - \gamma_y|}$
(3) $\gamma_x - \gamma_y$ (4) $\gamma_x + \gamma_y$

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Q19

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In a photoelectric effect experiment the threshold wavelength of light is 380 nm. If the wavelength of incident light is 260 nm, the maximum kinetic energy of emitted electrons will be:

$$\text{Given } E \text{ (in eV)} = \frac{1237}{\lambda \text{ (in nm)}}$$

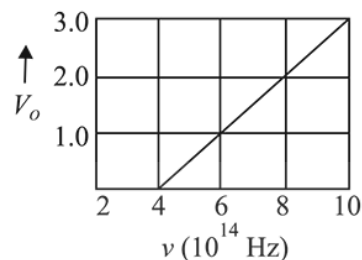
- (1) 1.5 eV (2) 3.0 eV (3) 4.5 eV (4) 15.1 eV

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Q20

The stopping potential V_0 (in volt) as a function of frequency (ν) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be :

(Given : Planck's constant (h) = 6.63×10^{-34} Js, electron charge $e = 1.6 \times 10^{-19}$ C)



- (1) 1.82 eV (2) 1.66 eV (3) 1.95 eV (4) 2.12 eV

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Experimental Physics - Answers

Q1

(3) Resistance, $R = \frac{\rho \ell}{A}$

$$R = \rho \frac{\ell}{A} \times \frac{\ell}{\ell} = \frac{\rho \ell^2}{V} \quad [\because \text{Volume (V)} = A\ell.]$$

Since resistivity and volume remains constant therefore
% change in resistance

$$\frac{\Delta R}{R} = \frac{2\Delta \ell}{\ell} = 2 \times (0.5) = 1\%$$

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Q2

(1) From Einstein's photoelectric equation,

$$\frac{hc}{\lambda_1} = \phi + \frac{1}{2} m (2v)^2 \quad \dots(i)$$

$$\text{and } \frac{hc}{\lambda_2} = \phi + \frac{1}{2} mv^2 \quad \dots(ii)$$

As per question, maximum speed of photoelectrons in two cases differ by a factor 2

From eqn. (i) & (ii)

$$\Rightarrow \frac{\frac{hc}{\lambda_1} - \phi}{\frac{hc}{\lambda_2} - \phi} = 4 \Rightarrow \frac{hc}{\lambda_1} - \phi = \frac{4hc}{\lambda_2} - 4\phi$$

$$\Rightarrow \frac{4hc}{\lambda_2} - \frac{hc}{\lambda_1} = 3\phi \Rightarrow \phi = \frac{1}{3} hc \left(\frac{4}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$= \frac{1}{3} \times 1240 \left(\frac{4 \times 350 - 540}{350 \times 540} \right) = 1.8 \text{ eV}$$

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Q3

(4) According to question, there are two EM waves with different frequency,

$$B_1 = B_0 \sin(\pi \times 10^7 c)t$$

$$\text{and } B_2 = B_0 \sin(2\pi \times 10^7 c)t$$

To get maximum kinetic energy we take the photon with higher frequency

$$\text{using, } B = B_0 \sin \omega t \text{ and } \omega = 2\pi v \Rightarrow v = \frac{\omega}{2\pi}$$

$$B_1 = B_0 \sin(\pi \times 10^7 c)t \Rightarrow v_1 = \frac{10^7}{2} \times c$$

$$B_2 = B_0 \sin(2\pi \times 10^7 c)t \Rightarrow v_2 = 10^7 c$$

where c is speed of light $c = 3 \times 10^8 \text{ m/s}$

Clearly, $v_2 > v_1$

so KE of photoelectron will be maximum for photon of higher energy.

$$v_2 = 10^7 c \text{ Hz}$$

$$h\nu = \phi + KE_{\text{max}}$$

energy of photon

$$E_{\text{ph}} = h\nu = 6.6 \times 10^{-34} \times 10^7 \times 3 \times 10^9$$

$$E_{\text{ph}} = 6.6 \times 3 \times 10^{-19} \text{ J}$$

$$= \frac{6.6 \times 3 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 12.375 \text{ eV}$$

$$KE_{\text{max}} = E_{\text{ph}} - \phi$$

$$= 12.375 - 4.7 = 7.675 \text{ eV} \approx 7.7 \text{ eV}$$

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Q4

(3) Least count of screw gauge,

$$LC = \frac{\text{Pitch}}{\text{No. of division}}$$

$$= 0.5 \times 10^{-3} = 0.5 \times 10^{-2} \text{ mm}$$

$$+ \text{ve error} = 3 \times 0.5 \times 10^{-2} \text{ mm}$$

$$= 1.5 \times 10^{-2} \text{ mm} = 0.015 \text{ mm}$$

$$\text{Reading} = \text{MSR} + \text{CSR} - (+\text{ve error})$$

$$= 5.5 \text{ mm} + (48 \times 0.5 \times 10^{-2}) - 0.015$$

$$= 5.5 + 0.24 - 0.015 = 5.725 \text{ mm}$$

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Q5

Experimental Physics

(4) Using, $\lambda = \frac{h}{p}$ {given: $\lambda = 7.5 \times 10^{-12}$ }

$$\Rightarrow P = \frac{h}{\lambda}$$

Minimum energy required,

$$KE = \frac{p^2}{2m} = \frac{(h/\lambda)^2}{2m} = \frac{\left\{ \frac{6.6 \times 10^{-34}}{7.5 \times 10^{-12}} \right\}^2}{2 \times 9.1 \times 10^{-31}}$$

$$J = 25 \text{ keV}$$

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Q6

(3)

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Q7

(3) using, intensity $I = \frac{nE}{At}$

n = no. of photoelectrons

$$\Rightarrow 16 \times 10^{-3} = \left(\frac{n}{t} \right) \times \frac{10 \times 1.6 \times 10^{-19}}{10^{-4}} \text{ or, } \frac{n}{t} = 10^{12}$$

So, effective number of photoelectrons ejected per unit time = $10^{12} \times 10/100 = 10^{11}$

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Q8

(4) de-Broglie wavelength,

$$\lambda = \frac{h}{mv} = 10^{-3} \left(\frac{3 \times 10^8}{6 \times 10^{14}} \right) \quad \left[\because \lambda = \frac{c}{\nu} \right]$$

$$\nu = \frac{6.63 \times 10^{-34} \times 6 \times 10^{14}}{9.1 \times 10^{-31} \times 3 \times 10^5}$$

$$\nu = 1.45 \times 10^6 \text{ m/s}$$

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Q9

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(3) Let ϕ = work function of the metal,

$$\frac{hc}{\lambda_1} = \phi + eV_1 \quad \dots\dots(i)$$

$$\frac{hc}{\lambda} = \phi + eV \quad \dots\dots(ii)$$

Subtracting (ii) from (i) we get

$$hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) = e(V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{hc}{e} \left(\frac{\lambda_2 - \lambda_1}{\lambda_1 \cdot \lambda_2} \right) \quad \left[\begin{array}{l} \lambda_1 = 300 \text{ nm} \\ \lambda_2 = 400 \text{ nm} \\ \frac{hc}{e} = 1240 \text{ nm} - V \end{array} \right]$$

$$= (1240 \text{ nm} - V) \left(\frac{100 \text{ nm}}{300 \text{ nm} \times 400 \text{ nm}} \right)$$

$$= 1.03 \text{ V} \approx 1 \text{ V}$$

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Q10

(2) Least count of main scale of screw gauge = 1 mm

Least count of screw gauge

$$= \frac{\text{Pitch}}{\text{Number of division on circular scale}}$$

$$5 \times 10^{-6} = \frac{10^{-3}}{N}$$

$$\Rightarrow N = 200$$

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Q11

Experimental Physics

- (3) de Broglie wavelength (λ) is given by
 $K = qV$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \quad (\because p = \sqrt{2mK})$$

Substituting the values we get

$$\begin{aligned} \therefore \frac{\lambda_A}{\lambda_B} &= \frac{\sqrt{2m_B q_B V_B}}{\sqrt{2m_A q_A V_A}} = \sqrt{\frac{4m.q.2500}{m.q.50}} \\ &= 2\sqrt{50} = 2 \times 7.07 = 14.14 \end{aligned}$$

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Q12

(BONUS)

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Q13

- (4) Using, wavelength, $\lambda = \frac{12375}{\Delta E}$

$$\text{or, } \lambda = \frac{12375}{4.9} \approx 250\text{nm}$$

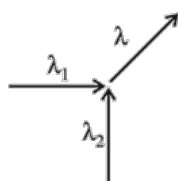
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Q14

- (1) From the de-Broglie relation,

$$p_1 = \frac{h}{\lambda_1}$$

$$p_2 = \frac{h}{\lambda_2}$$



Momentum of the final particle (p_f) is given by

$$\therefore p_f = \sqrt{p_1^2 + p_2^2}$$

$$\Rightarrow \frac{h}{\lambda} = \sqrt{\frac{h^2}{\lambda_1^2} + \frac{h^2}{\lambda_2^2}}$$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$$

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Q15

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- (4) We have

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\text{or } g = 4\pi^2 \frac{\ell}{T^2}$$

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta R}{Q} \times 100 + 2 \frac{\Delta T}{T} \times 100$$

$$= \frac{0.1}{55} \times 100 + 2 \left(\frac{1}{30} \right) \times 100$$

$$= 0.18 + 6.67 = 6.8\%$$

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Q16

$$(3) d = \frac{M}{V} = \frac{M}{L^3} = ML^{-3}$$

$$\frac{\Delta d}{d} = \frac{\Delta M}{M} + 3 \frac{\Delta L}{L}$$

$$= \frac{0.10}{10.00} + 3 \left(\frac{0.01}{0.10} \right) = 0.31 \text{kgm}^{-3}$$

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Q17

- (3) Here $\omega = 2\pi \times 6 \times 10^{14}$

$$\Rightarrow f = 6 \times 10^{14} \text{ Hz}$$

Wavelength

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^{14}} = 0.5 \times 10^{-6} \text{ m} = 5000 \text{ \AA}$$

$$\text{Given } E = \frac{12375}{5000} = 2.48 \text{ eV}$$

$$\text{Using } E = W + eV_s$$

$$\Rightarrow 2.48 = 2 + eV_s$$

$$\text{or } V_s = 0.48 \text{ V}$$

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Q18

$$(2) P_1 - P_2 = (P_1 + P_2) = P$$

$$\text{As } P \propto \frac{1}{\lambda}$$

$$\text{or } \frac{1}{\lambda_x} - \frac{1}{\lambda_y} = \frac{1}{\lambda}$$

$$\text{or } \frac{\lambda_y - \lambda_x}{\lambda_x \lambda_y} = \frac{1}{\lambda}$$

$$\therefore \lambda = \frac{\lambda_x \lambda_y}{|\lambda_y - \lambda_x|}$$

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Q19

$$(1) KE_{\max} = E - \phi_0$$

(where E = energy of incident light ϕ_0 = work function)

$$= \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$= 1237 \left[\frac{1}{260} - \frac{1}{380} \right]$$

$$= \frac{1237 \times 120}{380 \times 260} = 1.5 \text{ eV}$$

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Q20

$$(2) f_0 = 4 \times 10^{14} \text{ Hz}$$

$$W_0 = hf_0 = 6.63 \times 10^{-34} \times (4 \times 10^{14}) \text{ J}$$

$$= \frac{(6.63 \times 10^{-34}) \times (4 \times 10^{14})}{1.6 \times 10^{-19}}$$

$$= 1.66 \text{ eV}$$

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